

Postfire lichen—spruce woodland recovery at the limit of the boreal forest in northern Quebec

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A 250-year postfire plant chronosequence in well-drained sites at the northern limit of the Boreal Forest in the Grande rivière de la Baleine area, northern Quebec, was reconstructed from nine sites associated with the development of the lichen—spruce woodland. Most species recorded along the chronosequence reinvaded burned sites within 15 years after fire, whereas important vegetational changes occurred during the first 100 years of postfire recovery, corresponding to full development of the *Cladina stellaris*—spruce woodland. No vascular plant species replacement was observed during succession, whereas by contrast a well-defined lichen—bryophyte sequence occurred along five successional stages. Species diversity (Shannon index) was low 4 years after fire, but it has reached a maximum about 25 years after fire and then dropped and stabilized at a low value in old-growth woodlands dominated by black spruce (*Picea mariana* (Mill.) BSP) and *C. stellaris*. The active period of black spruce sexual regeneration spans about 20–25 years, with maximum regeneration occurring between 5 and 14 years after fire. After this period, seed regeneration is mostly sporadic and sustained layering becomes fairly common at all sites. From 100 to 250 years after fire, no significant changes were observed in vegetation structure, floristic composition, species diversity, and spruce regeneration, suggesting that lichen woodlands are self-perpetuating in absence of fire. In limiting spruce regeneration at the ground surface, the lichen cover seems to be the most important factor controlling the open structure of the lichen—spruce woodland.

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La chronoséquence végétale après-feu menant à la pessière à lichens dans la région de la Grande rivière de la Baleine, située dans la partie nord de la forêt boréale au Québec, a été reconstituée par l'étude de neuf sites bien drainés. La plupart des espèces de la chronoséquence recolonisent les sites brûlés au cours des 15 années suivant l'incendie, alors que d'importants changements de végétation se produisent jusque vers 100 ans après-feu, soit jusqu'au développement d'une pessière ouverte à *Cladina stellaris*. Il n'y a pas de remplacement d'espèces chez les plantes vasculaires dominantes au cours de la succession. Par contre, une séquence bien définie d'espèces caractérise le développement des strates lichénique et muscinale en fonction de cinq stades successionnels. La diversité spécifique (indice de Shannon), faible 4 ans après-feu, atteint un maximum après environ 25 ans, diminue, puis se stabilise à une faible valeur dans les vieilles forêts dominées par l'épinette noire (*Picea mariana* (Mill.) BSP) et *C. stellaris*. L'épinette noire connaît une période de régénération sexuée active débutant immédiatement après-feu et s'étendant sur 20–25 ans, avec un maximum de régénération entre 5 et 14 ans après-feu. Après cette période, la régénération par graines devient sporadique et le maintien de l'épinette noire s'effectue principalement par voie végétative (marcottage). Entre 100 et 250 ans après-feu, la structure de la végétation, la composition floristique, la diversité, le mode de régénération de l'épinette noire et son recouvrement ne présentent pratiquement plus de variations, suggérant que la pessière à lichens se perpétue en absence de feu. En limitant la régénération sexuée et l'expansion de l'épinette noire, le couvert de lichens apparaît comme le principal facteur responsable de la structure ouverte et du maintien à long terme des pessières à lichens.

Introduction

In boreal forest ecosystems, fire is one of the most important factors that controls floristic composition, vegetation structure, and regional vegetation mosaic along a successional gradient (Heinselman 1981). Most studies have shown that no tree species replacement is observed during succession in postfire boreal ecosystems (Dix and Swan 1971; Carleton and Maycock 1978; Cogbill 1985) where the tree canopy is mostly monospecific. Lichen—spruce woodlands extend throughout the Canadian Boreal Forest (Rowe 1972), with distinct postfire vegetational types dominated by lichens and bryophytes. Several postfire chronosequences describing the vegetational development of the lichen—spruce woodland were reported for the Northwest Territories (Maikawa and Kershaw 1976; Black and Bliss 1978; Johnson 1981), northern Saskatchewan (Scotter 1964), southeastern Labrador (Foster 1985), and Newfoundland (Bergerud 1971). According to these studies, the lichen—moss succession may be summarized as follows: the ground surface is first colonized by moss species and crustaceous lichens, followed by several *Cladonia* species, and, generally in less than 100 years, *Cladina stellaris* or

Stereocaulon paschale, according to area (Kershaw 1977), form a continuous cover in mature forests. Foster (1985) and Maikawa and Kershaw (1976) suggested that the lichen cover was eventually replaced by a moss cover as the tree canopy closed in absence of fire during a long period of time (e.g., 150 years). The lichen—spruce woodland is thus considered as a successional stage leading to the development of the feather moss closed-crown forest. However, contrasting views indicate that the lichen—spruce woodland was self-perpetuated during the past 800–900 years at the treeline in northern Quebec (Payette *et al.* 1985), whereas Strang (1973) argued that upland woodlands along the Mackenzie River, Northwest Territories, shift to tundra vegetation in absence of fire.

The northern part of the Boreal Forest, in the Grande rivière de la Baleine area (Fig. 1), is dominated by a fire-induced vegetation mosaic (Payette *et al.* 1989). On well-drained sites, several postfire successional stages with successful forest regeneration occur in association with mature open forests or woodlands of the *Cladina stellaris*—*Picea mariana* type. The objective of our study is to analyze the regional postfire chronosequence at these sites in considering a long time

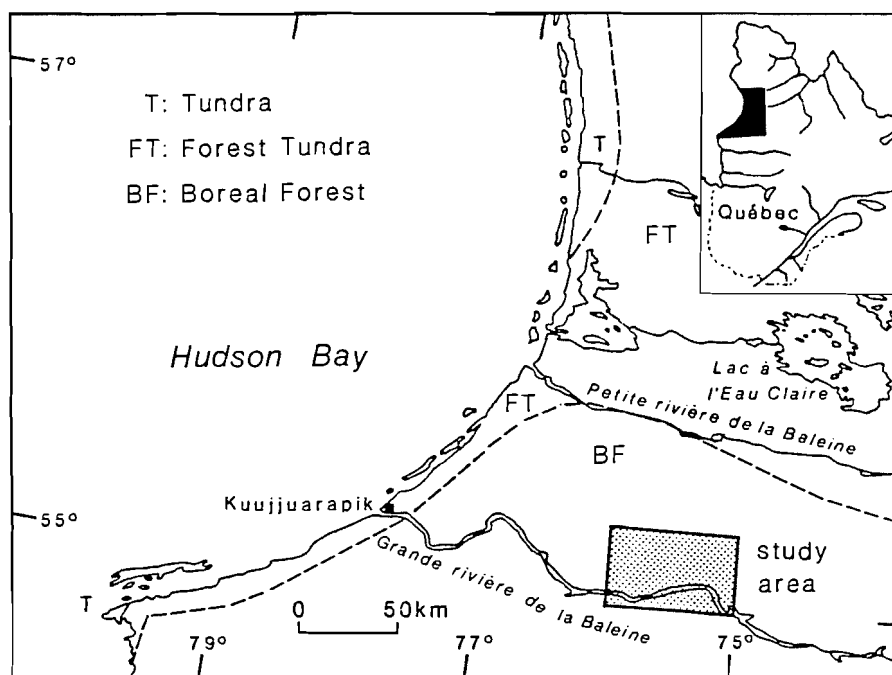


FIG. 1. Location of the study area.

sequence for the development and fate of the lichen-spruce woodland. This development can be addressed from two perspectives. (i) the tree canopy remains open during succession and the lichen-spruce woodland self-perpetuates or (ii) the gradual expansion of the tree canopy results in the development of the feather moss closed-crown forest. Progressive shading as the tree canopy closed favors moss expansion at the expense of intolerant lichens. These perspectives discard the possibility of removal of tree canopy as succession proceeds because a sustained postfire forest regeneration is currently occurring in the area.

Because there is no published record on postfire plant chronosequences in northern Quebec, we have made a detailed analysis of postfire succession in well-drained sites leading to the development of the lichen-spruce woodland. The reconstruction of the postfire chronosequence was made possible by the study of nine sites where climate and soil conditions were considered homogeneous. Time after fire was considered the prime factor determining the pattern of vegetation recovery. Floristic and edaphic changes were analyzed from large quadrats in relation to the temporal development of coniferous populations dominated by black spruce. This part of the study was based on the age structure of spruce populations in order to evaluate the role of sexual and vegetative regeneration during succession. Although this topic has been neglected in most studies on postfire chronosequences, we think that it is a major feature both for describing and explaining the pattern of postfire regeneration and the long-term development of the lichen-spruce woodland in absence of fire.

Study area

The study sites are located north and south of the Grande rivière de la Baleine, about 100 km east of the Kuujuaupik settlement along the Hudson Bay coast, between 55–55°20'N and 75–76° W (Fig. 1). The area lies within the Canadian Precambrian Shield dominated by Archean granitic and gneissic rocks (Eade 1966). According to recent data, the area was deglaciated about 7000 years ago (Dyke and Prest

1987), and postglacial marine transgression did not invade lowland sites. The deep valley of the Grande rivière de la Baleine is bordered by a rolling plateau with rocky hills underlain by a thin sheet of till deposits. Mean elevation ranges between 250 and 350 m above sea level. The estimated mean annual temperature is -4°C ; the mean July temperature, 11°C ; and the mean January temperature, -24°C . The mean annual precipitation is about 700 mm, with 40% as snow. The mean frost-free period does not exceed 60–70 days (Wilson 1971).

The regional vegetation corresponds to the northern part of the Boreal Forest, where the forest cover is continuous on well-drained sites (Payette 1983). The northern limit of the Boreal Forest coincides with the area where natural openings occupied by tundra vegetation are found mostly on hilltops. These sites belong to the southern forest-tundra. Macrofossil evidence suggests that they were previously forested, but postfire spruce regeneration has been deficient sometime in the past (Payette and Gagnon 1985).

Black spruce is the dominant forest species in the study area, while tamarack (*Larix laricina*) is particularly abundant in humid sites. Several tree species (*Pinus banksiana*, *Betula papyrifera*, and *Populus tremuloides*) are also found along the Grande rivière de la Baleine valley where they reach their northern range limit. Balsam poplar (*Populus balsamifera*) clones are found mostly along brooks and ravines and extend deep into the forest-tundra. Lichen-spruce woodlands are the most representative forest types in this part of the northern Boreal Forest. The woodlands are growing on well-drained soils and present a characteristic open structure of spruce and shrub species underlain by a continuous lichen cover. Feather moss forests are located in moist sites, whereas treeless peatlands are scattered throughout the area.

The recent fire history of the northern part of the Boreal Forest, including the Grande rivière de la Baleine area, was outlined by Payette *et al.* (1989). It was shown that 65% of this boreal region burned during the 1920–1984 period. The natural fire rotation period (Heinselman 1973) was estimated to about 100 years. Data on postfire regeneration in the area indicate normal restocking of spruce populations at most sites.

Methods

Selection of the study sites was based on fire data (fire-scar dates and areal extent of individual fires) available for this part of the

TABLE 1. Physical characteristics of the study sites

	Site								
	1	2	3	4	5	6	7	8	9
Postfire age (years)	4	14	23	38	42	65	129	ca.250	ca.250
Site location (lat., long.)	55°17'35"N, 75°45'00"W	55°12'40"N, 75°03'30"W	55°08'15"N, 75°37'05"W	55°14'25"N, 75°26'55"W	55°06'45"N, 75°30'30"W	55°09'25"N, 75°45'05"W	55°00'45"N, 75°51'15"W	55°17'55"N, 75°51'45"W	55°18'10"N, 75°52'35"W
Altitude (m)	300	300	280	290	280	310	310	280	280
Slope (%)	3	3	0	3	7	1	3	7	10
Texture									
B horizon									
Sand (%)	68	75	78	68	84	75	77	73	83
Silt (%)	30	23	19	30	14	23	21	25	15
Clay (%)	2	2	3	2	2	2	2	2	2
C horizon									
Sand (%)	62	34	76	60	73	64	78	62	78
Silt (%)	36	63	21	37	26	33	19	35	19
Clay (%)	2	2	3	3	1	3	3	2	3
Sesquioxides (B horizon)									
Fe (%)	0.08	0.51	0.26	0.04	0.09	0.11	0.22	0.25	0.33
Al (%)	0.24	0.51	0.55	0.32	0.37	0.27	0.40	0.66	0.60

northern Boreal Forest (Payette *et al.* 1989). The postfire ages of the selected sites were 4, 14, 23, 38, 42, 65, 129, and ca. 250 years. These sites were chosen in relation to the age span of all postfire stages currently observed in the area. Close time intervals between young postfire sites were indicated because many ecological changes are occurring early during succession, whereas only minor changes occur afterwards. In absence of fire scar, the age of the two oldest woodlands (sites 8 and 9) was obtained from age-structure surveys and radiocarbon (^{14}C) dating. Charcoal sampled at site 8 gave a ^{14}C age of 360 ± 70 BP (UL-309), and 440 ± 70 BP (UL-310) at site 9. Since fire dating from age structure is more precise than ^{14}C dating, postfire stand age determination from this technique was considered reliable, whereas the ^{14}C dates correspond to approximate age of burned spruces.

Homogeneity of edaphic conditions between selected sites was considered important to have time as the active factor of vegetation development during succession. Therefore, all the study sites were situated on well-drained till with a low angle. The topography of these sites was smooth, drumlin-like, with a uniform vegetation cover over a large area. Accessibility to the sites with a floatplane was of prime importance in site selection. The physical characteristics of the study sites are briefly described in Table 1.

Each site was sampled within a 20×50 m quadrat with the long axis parallel to the dominant topographical features, most often in an east-west direction. The percent cover of all vegetation strata has been estimated from 10 subquadrats of 100 m^2 (10×10 m), using 11 cover categories (+; < 1; 1, 1–10; 2, 10–20; 3, 20–30; ... 10, 90–100). The mean of the median values of all cover categories was used to evaluate areal coverage of each stratum at a particular site. The species abundance was calculated from linear relevés. Twenty 1-m long lines, parallel to the long axis of the quadrat, were randomly positioned within the 1000 m^2 quadrat. The species cover was measured along each line from 10-cm long segments using the same cover categories as cited above. The species cover at a given site was evaluated also from the mean of the median values of all cover categories (species) in the 200 linear relevés per site. The species frequency was calculated as the ratio of the number of 10-cm long segments where the species was present to the total number of segments. Spruce cover was obtained by the linear relevés and the subquadrat stratum sampling. Only the cover values of black spruce from the subquadrat stratum sampling will be considered thereafter because the technique of the linear relevé with small segments is best suited for cover and frequency evaluation of small plants such as lichens and mosses. Botanical nomenclature was based on Morisset *et al.* (1987) for vascular plants, Ireland *et al.* (1980) for mosses, Egan (1987) for lichens, and Stotler and Crandall-Stotler (1977) for liverworts.

An index of species diversity was calculated for each site. Diversity is a measure of species composition of a stand in terms of species number and relative abundance (Legendre and Legendre 1984). It varies according to species richness and regularity. Thus, for the same number of species, the diversity index is maximal when all species are equally represented (maximal regularity) and minimal when one species is largely dominant, the other species being represented by one individual (minimal regularity). The index value also increases with the number of species. The Shannon index H' , which respects the above criteria, was used and is defined as

$$H' = \sum p_i \log_e p_i$$

where p_i corresponds to the cover proportion of species i in comparison with the total coverage of the n species. The measure of regularity was obtained in comparing the calculated diversity to the maximum value that the latter would take if all the species were equally represented. Regularity is thus equal to $R = H'/H'_{\max} = H'/\log_e n$, where n is the number of species. The number of species registered in the linear relevés corresponds to the measure of richness at a given site.

The origin (seed or layer), height (according to six classes: 1, 0–0.3 m; 2, 0.3–1.4 m; 3, 1.4–2.5 m; 4, 2.5–5 m; 5, 5–10 m; 6, > 10 m), diameter at breast height (i.e., 1.4 m above the ground),

TABLE 2. Relationship between age and height of black spruce for age correction

Site	Height classes (m)	Linear regressions ($Y = \text{age}$, $X = \text{height in cm}$)	r	SE	Age prediction at 30 cm
4	0.3–2.5	$Y = 0.3 + 0.33X$	0.93	2.0	10
	2.5–5.0	$Y = 0.3 + 0.23X$	0.81	2.6	7
5	0.3–2.5	$Y = 0.41X$	0.94	2.5	12
	2.5–5.0	$Y = -0.1 + 0.25X$	0.90	1.9	7
6	0.3–2.5	$Y = 0.1 + 0.46X$	0.88	3.9	14
	2.5–5.0	$Y = 0.5 + 0.28X$	0.89	2.5	9
	5.0–10.00	$Y = 0.19X$	0.80	2.2	6
7	>5.0	$Y = 1.4 + 0.18X$	0.94	2.1	7
	Layers	$Y = 0.34X$	0.88	2.8	10
8	5.0–10.0	$Y = 0.3 + 0.50X$	0.95	5.6	15
	Layers	$Y = -0.1 + 0.56X$	0.97	4.6	17

and age of all tree stems recorded in the 1000-m² quadrat were determined systematically. Leaning branches at the soil surface with adventitious roots were identified as layers. Stem age was determined from tree-ring cores showing the central pith sampled at 30 cm above the ground. Smaller stems were sampled at the soil surface. In old-growth woodlands, at least 100 layers less than 1.4 m high were randomly sampled in 100-m² subquadrats. This procedure necessitated a 700-m² surface in site 7, 1 200-m² surface in site 8, and a 100-m² surface in site 9. The tree-ring cores and sections were sanded and examined under the binocular microscope.

An age correction was used on dated cores for determination of the real age of spruce stems sampled at 30 cm above the ground. A specific age correction was applied to each site. Trees of different size may have different growth rates, and use of the same correction factor could produce unrealistic age structure. To obviate the problem when age correction was needed at a given site, five individuals per height class (height classes: 1, 0.3–2.5 m; 2, 2.5–5 m; 3, 5–10 m; 4, >10 m) were randomly sampled. On each sampled tree, sections were obtained at 0, 10, 20, 30, 50, and 100 cm above the ground. A linear regression was used with age–height data on samples of each height class (Table 2). Except for site 7, all the curve slopes were found to be significantly different (at $P = 0.05$), suggesting that among height classes at a given site conifers had different growth rates. Therefore, the age correction on dated cores was applied according to the height class of individuals using regression curves. At site 7, the data were grouped and only one age correction was used for individuals >5 m. Six or seven years, depending on particular conditions, were added to stem age at 30 cm above the ground for all individuals of the highest class per site except in site 8 (Table 2). In sites 7 and 8, many layers were cored at 30 cm above the soil surface, and a similar age correction was used but without considering the height class. In sites where the age of all individuals was determined at the collar, age structure was based on a 2-year class (sites 2 and 3). A 5-year class was used at the other sites to minimize error associated with age determination at 30 cm above the ground.

At all study sites, the soil profile was strongly podzolized, with a bleached horizon 2–3 cm thick and sesquioxide accumulation in the B horizon (Table 1). Sampling was based on the dominance of major horizons (FH, B, and C) in each podzolic profile. The thickness of FH horizon was measured at 10 points randomly positioned within the 1000 m² quadrat. In addition, thickness of the lichen carpet, corresponding to the mean height of the tallest podetia, was evaluated at these points. In old burned sites, the thallus base was partially decomposed, thus only the height of the living part was measured. A sample of FH, B, and C horizons were taken randomly at each site. Texture of mineral samples (<2 mm) was determined for verification of parent material homogeneity, pH, organic matter content, total N, exchangeable bases (Ca, Mg, K, Na), and cationic exchange capacity were also measured. Laboratory analyses were conducted according to procedures outlined by McKeague (1978).

Results

Plant chronosequence

The overall development of the lichen–spruce woodland during succession is first apparent from changes in vegetation structure. Most changes occurred during the first 65 years after fire, and at times vegetation structure remained virtually unchanged even 250 years after fire (Fig. 2). After initial lichen colonization, subsequent changes were associated with development of the tree stratum. Only the shrub stratum was particularly important 4 years after fire. It reached a maximum cover of 30% at year 40, with growing spruces as a dominant element of the shrub population. The lichen stratum expanded rapidly a few years after fire and dominated throughout the sequence with 90% cover. The tree stratum was first noticed at year 65, but it did not exceed 20% cover. Moss and herb cover remained very low throughout the sequence.

Vegetation development during succession indicated no apparent replacement in vascular species. The main shrub species in mature forests, such as *Betula glandulosa*, *Ledum groenlandicum*, and *Vaccinium angustifolium*, reproduced vegetatively from subterranean organs that escaped fire. New shoots are usually formed a few days or weeks after fire. Although the abundance of shrub species varied between sites, it was generally lower in old stands, with a maximum cover 20 to 40 years after fire, as shown by *B. glandulosa*. Black spruce has the capacity to rapidly reinvade burned sites. The species cover gradually increased from the initial colonization period until 65 years after fire, when it has reached a cover of about 20%, a value that remained fairly constant for the rest of the chronosequence (Fig. 3).

In contrast with vascular plants, a progressive change in floristic composition of lichen and moss vegetation was observed during succession. The development through time of the most characteristic species is shown in Fig. 3. During the first years of recovery, the burned surface was colonized mostly by vascular plants that can regenerate vegetatively. Only a few pioneer species, *Polytrichum juniperinum*, *Polytrichum piliferum*, and *Ceratodon purpureus*, were present 4 years after fire on mineral substrata. Later, the crustaceous lichen *Trapeziopsis granulosa* formed a dense crust at the soil surface, with maximum development at year 14 (Fig. 3c), whereas *Polytrichum* species followed the same colonizing pattern (Figs. 3a, 3b). Several *Cladonia* and *Cladina* species were present 14 years after fire, but they became dominant afterwards. Twenty-five years after fire, the vegetation cover was dominated by

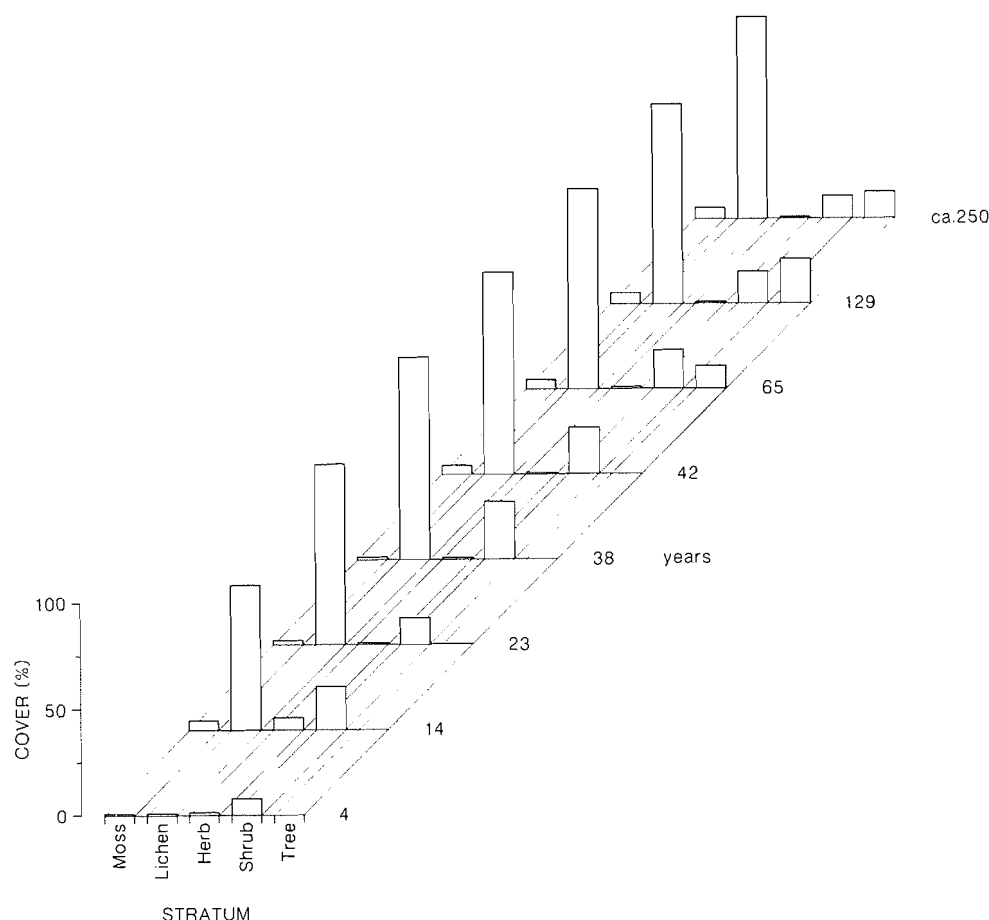


FIG. 2. Changes in vegetation structure during succession.

many *Cladonia* species, such as *C. macrophylla*, *C. deformis*, *C. sulphurina*, *C. coccifera*, *C. cornuta*, *C. crispata*, and *C. uncialis*. At this stage, all the species were very frequent but spatially restricted (Fig. 3). Then, a continuous lichen carpet formed at year 38 when fast-growing *Cladina mitis* populations achieved dominance (Fig. 3k), with *Cladonia uncialis* and *Cladonia crispata* as secondary species. At the time *Cladina mitis* reached maximum coverage (40%) 65 years after fire, *Cladina stellaris* increased in importance and *Cladonia* species were confined to marginal microhabitats, such as burned logs and small depressions between lichen polygons (Fig. 3n). The spatial development of *Cladina stellaris* populations showed that the species cover (80%) was fairly continuous 130 years after fire (Fig. 3m) and remained rather constant during the rest of the chronosequence, with *Cladina mitis* and *Cladina rangiferina* as secondary species (about 5% cover).

Thickness of the lichen carpet increased gradually from the start of the *Cladonia* expansion to 130 years after fire, when it reached about 12–13 cm (Fig. 4). However, as time passed, apical growth of the lichen thalli (*Cladina* spp.) was overcome to some extent by progressive death of the basal podetia. This explains why thickness of living lichen mats remained fairly constant until 250 years after fire.

Pleurozium schreberi and *Ptilidium ciliare* were the most characteristic bryophytes in mature forests. *Pleurozium schreberi* was first noticed under the shrub cover in 38- and 42-year-old stands. The species was associated with large black spruce clones in old-growth woodlands along with *Pti. ciliare*, which

had also the ability to coexist with *Cladina stellaris* in the lichen carpet. On the other hand, the most important herb species, *Deschampsia flexuosa*, *Oryzopsis pungens*, and *Cornus canadensis*, presented a very low score throughout the chronosequence, although they showed maximum development about 15 years after fire.

Changes in floristic composition and plant cover during the postfire chronosequence allowed us to identify broad successional stages, particularly on the basis of spatiotemporal development of lichen and bryophytes species. The first stage corresponds to the first years of recovery when the ground surface was mostly devoid of vegetation. The *T. granulosa* – *Polytrichum* spp. stage occurred about 15 years after fire and was followed by the *Cladonia* spp. stage, which climaxed at year 25. The *Cladina mitis* stage lasted for about 50 to 70 years; the first part of the stage began 30–40 years after fire with *Cladonia* species (*C. crispata* and *C. uncialis*) and terminated with *Cladina stellaris*. The *Cladina stellaris* stage, with accompanying species such as *C. mitis* and *C. rangiferina*, corresponded to the development of mature lichen–spruce woodlands.

During the chronosequence, the diversity index H' shifted rapidly from a low value a few years after fire to a maximum value 25 years after fire (Fig. 5a). Then, it decreased rapidly until year 65 and stabilized afterwards. The species richness appeared to follow the same trend, and regularity decreased from 4 to 250 years after fire (Fig. 5b). The low diversity index at year 4 was associated with a low number of species. Maximum diversity, species richness, and regularity coincided

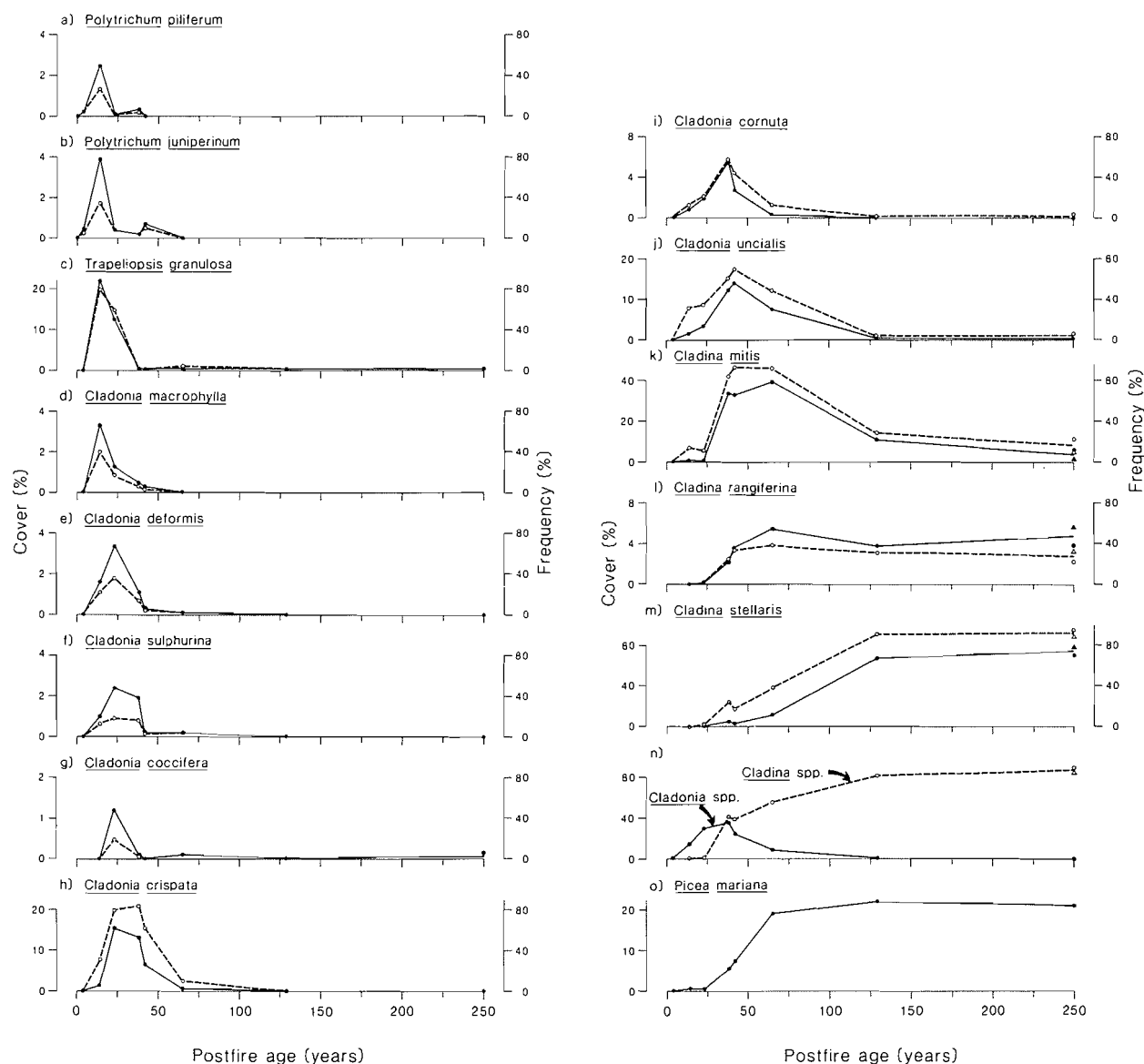


FIG. 3. Changes in cover (—) and frequency (---) of selected species during succession.

with the *Cladonia spp.* stage. Because the number of species remained fairly stable after year 40, decrease of H' at the end of the sequence may be related to low regularity caused by the dominance of *Cladina stellaris* and *Picea mariana*.

Black spruce development

At sites 1, 2, and 3 where the real age (root collar) of individuals was determined, spruce regeneration began 1–3 years after fire. Except in the 250-year-old woodland, mortality of standing spruces has rarely been observed. No dead spruce was reported at sites 1–6, and only three spruces died at site 7. At site 1, a 1982 burn, only three seedlings were recorded in 1986 (two germinated in 1984 and one in 1986). The age structure of the remaining sites (Fig. 6) indicated that black spruce regenerates sexually after fire during an active period of 20–25 years. Most individuals established during the first 15 years of the active period, with maximum colonization occurring 5 to 14 years after fire. After this period, sexual reproduction became rather sporadic (Fig. 6, sites 7 and 8).

Although this regeneration pattern may be generalized to

most postfire situations on well-drained soils, some sites were characterized by very dense postfire spruce populations, as in site 3 where 896 individuals (including a significant proportion of tamarack stems) were recorded in the 1000-m² quadrat (Fig. 6). At this site, spruce regeneration also peaked 15 years after fire. On the other hand, a second period of regeneration 30–34 years after fire was recorded at site 5 (Fig. 6). Finally, the period of maximum regeneration between 15 and 19 years after fire at site 7 was slightly delayed in comparison with that measured at the other sites.

Several dominant spruces at site 8 (last fire at ca. 1735) died during this century. Most dead spruces were located at the center of large clones, suggesting that they were once part of the postfire cohort. Although tree-ring sections from dead stems can be dated using the light-ring chronology developed in the study area (Filion *et al.* 1986), it has not been possible to date the exact time of spruce establishment because of rotten wood at the base of the trunks; hence, these individuals were not considered in the age structure. Therefore, age structure obtained from living spruces only cannot give an accurate pic-

TABLE 3. Density of black spruce stems (from seed origin and layering) at selected successional sites

Site	Postfire age (years)	Seed-origin stems ^a /1000 m ²	No. of layers/1000 m ²	Seed-origin stems producing layers (%)	Mean no. of layers/clone ($\bar{X} \pm \text{SE}$)
1	4	3	—	—	—
2	14	28	—	—	—
3	23	896	—	—	—
4	38	73	—	—	—
5	42	146	5	3.4	1 $\pm$ 0
6	65	145	32	6.9	3.2 $\pm$ 3.1
7	129	70	1114	71.4	23.1 $\pm$ 23.7
8	ca.250	45	1109	60.0	39.2 $\pm$ 38.2

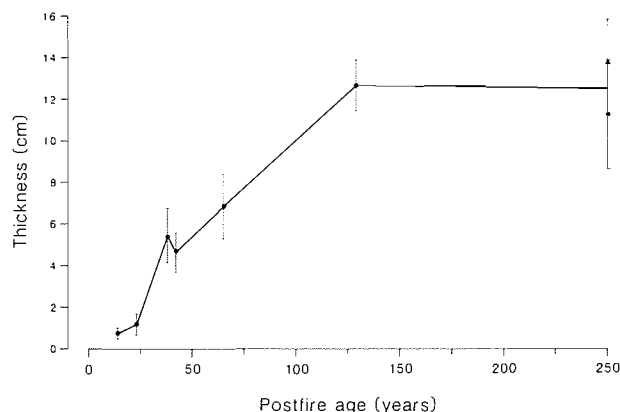
^aIncludes living and dead stems.

FIG. 4. Changes in thickness of the lichen mat during succession. Vertical bars indicate standard deviation.

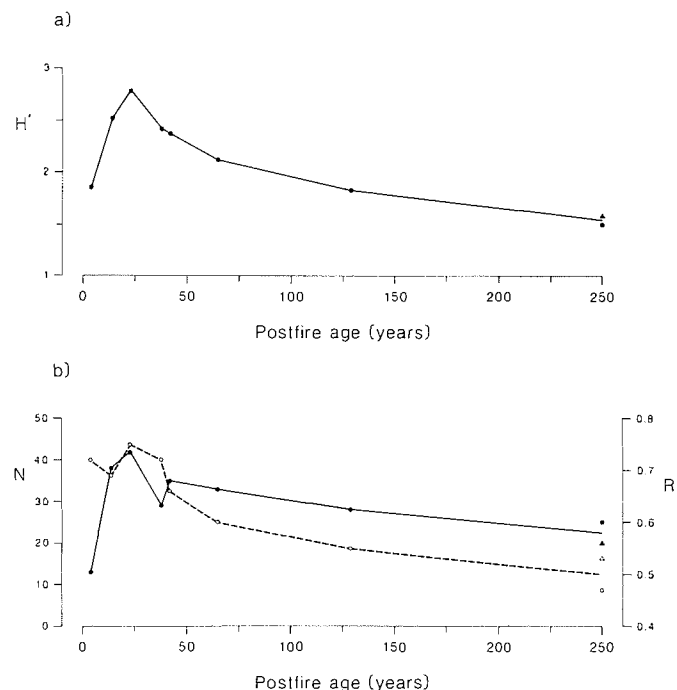
ture of site history (Fig. 6). It does indicate, however, that spruce regenerated sexually, although sporadically, during a long fire-free period. Vegetative regeneration by layering took over sexual regeneration in old-growth lichen woodlands (Fig. 6, sites 7 and 8). First recorded in a 42-year-old stand, layering became the dominant, regenerative process in old-growth lichen woodland (Fig. 6, Table 3). Young spruces can produce layers early during succession, but it was only in old-growth forests that most spruces regenerated vegetatively (Table 3). In old stands, the number of layers per clone varied largely, from 1 to more than 100; age and size of layers appeared to follow the same trend.

Soil conditions

In most sites, the FH horizon was not well developed (1–3 cm), and horizon thickness did not change much during succession (Fig. 7). Also, no particular trend was depicted in chemical soil conditions during succession (Fig. 8). Conversely, changes in pH, cationic exchange capacity, nitrogen content, and exchangeable cations were closely associated with organic matter content. At each site, there were large variations in the value of all these factors between organic and mineral horizons. Finally, pH horizon from one site to the other, as well as the overall low organic content, remained rather constant during succession.

Discussion

Our results suggest that the overall pattern of position vegetation recovery was initiated early during succession. Most

FIG. 5. Changes in (a) diversity (H') and (b) richness (N ;—) and regularity (R ;---) during succession.

species established during the first years after fire, a vegetational development substantiating Egler's concept of initial floristic composition (Finegan 1984). By contrast, it differed from the recovery sequence leading to the *Ple. schreberi*—spruce forest, as reported by Foster (1985), where tree regeneration (*Picea mariana* and *Abies balsamea*) started immediately after fire and proceeded gradually until replacement of the lichen—spruce woodland by the closed-crown feather moss forest.

Burned, well-drained sites of the Grande rivière de la Baleine area were colonized by vascular species immediately after fire, whereas most lichen species established during the first 15 years of postfire recovery. Only a few species were recorded 15 years after fire, and the most important vegetational changes during the first 100 years were associated with changes in species abundance.

The early colonization of vascular plants was an indication of their adaptation to recurrent fires, particularly the shrub species able to resprout rapidly from subterranean organs that sur-

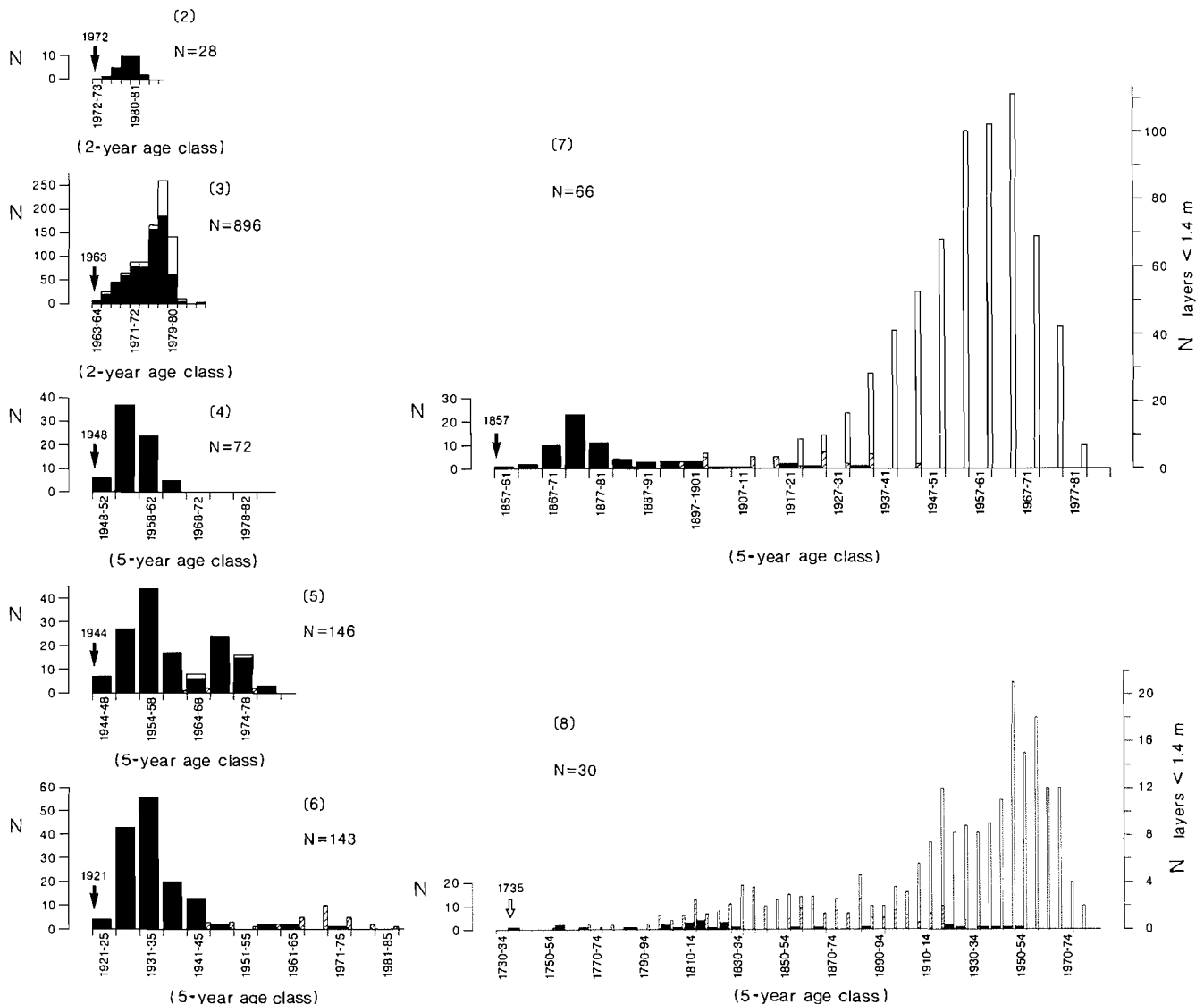


FIG. 6. Age structure of conifer populations during succession. ➔, fire date from fire-scar dating; ⇨, estimated date of last fire; ■, black spruce; □, tamarack; ▨, layers sampled in 1000 m² quadrats; □, subsamples of layers (< 1.4 m).

vived the last fire in the organic mat. Success of postfire shrub populations was controlled to a large extent of abundance of prefire species and fire severity. The uniform thickness of the FH horizon during succession (Fig. 7) suggested that light fires prevailed at these sites. In the Schefferville area, Auclair (1985) reported a 6% shrub mortality associated with fire conditions similar to that found in the Grande rivière de la Baleine area.

Generally, fire caused the death of the lichen thallus, and species recovery proceeded mostly from nearby diaspores along a well-defined sequence. A broad pattern of lichen succession applicable to most postfire development of lichen-spruce woodlands in Canada can be demonstrated from several studies (Scotter 1964; Bergerud 1971; Maikawa and Kershaw 1976; Black and Bliss 1978; Johnson 1981; Foster 1985). Regional differences do exist, however, as for example the lichen-spruce woodlands of the Abitau Lake, Northwest Territories, dominated by *S. paschale* (Maikawa and Kershaw 1976) and the eastern woodlands dominated by *Cladina stellaris*. Near Inuvik, Northwest Territories, Black and Bliss

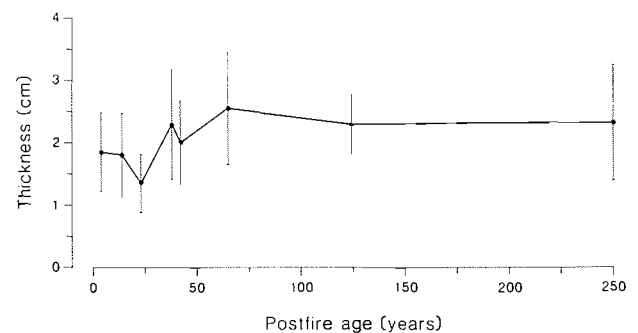


FIG. 7. Thickness of FH horizon during succession. Vertical bars indicate standard deviation.

(1978) outlined a postfire chronosequence where development of the lichen cover lasted about 200 years, in comparison with 100 years observed elsewhere. On the basis of floristic composition and rapidity of recovery, the lichen succession

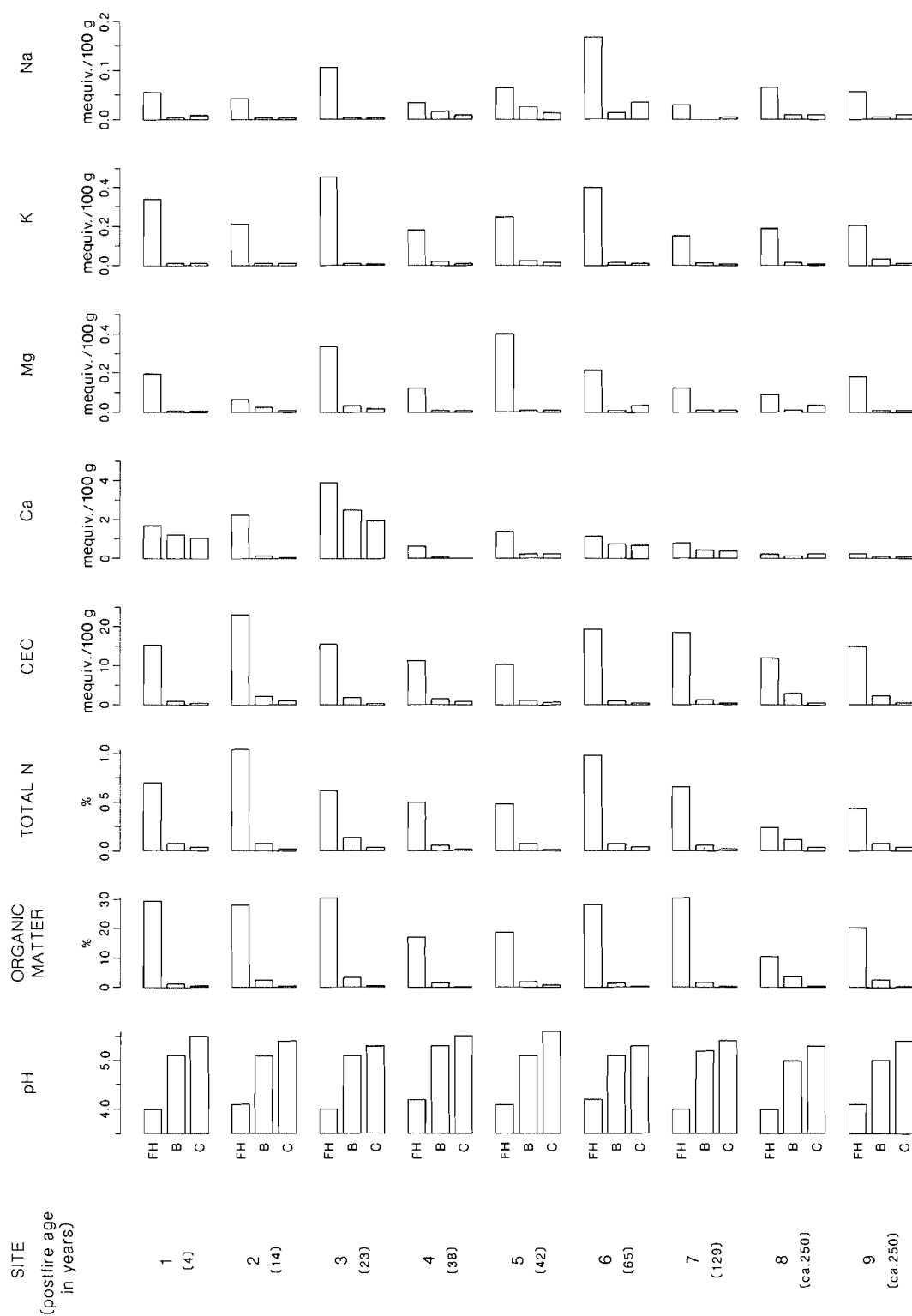


Fig. 8. Chemical analyses of FH, B, and C horizons at selected successional sites. CEC, cationic exchange capacity.

reported in our study is somewhat similar to those of other eastern Canadian regions (Bergerud 1971; Foster 1985). In southeastern Labrador however, lichens were displaced by mosses in late successional stages (Foster 1985).

In most studies (Ahti 1959; Bergerud 1971; Foster 1985; this study), lichen colonization was shown to take place several years after fire. Our results indicated a delay of 5 to 10–15 years between fire and lichen colonization, whatever the species. Although lichens were able to colonize burned sites during the same period of time, lichen succession appeared to be the result of success of propagule establishment and differential growth rates. The overall development of the moss stratum seemed to conform to the facilitation model (Connell and Slatyer 1977). During succession, the colonizing species (*Polytrichum* spp.) were replaced by other bryophytes (*Ple. schreberi* and *Pti. ciliare*) whose development was closely associated with expansion of the tree stratum several decades after fire.

The decreasing trend in species diversity, which started rather early during succession, was caused both by spatial development of black spruce and superior competitiveness of *Cladina* species over most other cryptogamic species. Maximal diversity occurred at the time the fruticulous lichen cover was still discontinuous, i.e., when competition for space was low. Then, a few decades after fire, the development of *Cladina* species influenced species abundance. Height (10–15 cm), growth form, and clonal expansion of *Cladina stellaris* (Yarranton 1975) have conferred to the species a competitive advantage over most other lichen species. The spatial expansion of *Cladina stellaris* reduced considerably the development of both *Cladina mitis* and *Cladina rangiferina* populations, as well as that of other lichen species. Therefore, lichen species exclusion by fast spatial growth of *Cladina stellaris* (and also *Cladina mitis*) was responsible for a decreasing diversity in late successional stages. This general pattern in species diversity has been also reported in other secondary successions (Loucks 1970; Auclair and Goff 1971; Shafi and Yarranton 1973; Oksanen 1986). Using lichen volumes as indicators of particular successional stages, Oksanen (1986) attributed the decreasing diversity and regularity observed in Finnish Scotch Pine forests during succession to spatial expansion of *Cladina* species.

Black spruce development

The characteristically fast postfire recovery of black spruce has been demonstrated in several studies (Ahlgren 1959; Hatcher 1963; Wein 1975; Viereck and Foote 1979; Black and Bliss 1980; Cogbill 1985), although the period of sustained sexual regeneration for this species is rather short (Sirois 1988; this study). Black spruce is well adapted to fire because of semi-serotinous cones that act as effective seed banks and gradually disperse the seeds many years after fire (Wilton 1963; Vincent 1965; Zasada *et al.* 1979). According to Viereck (1983), cones from burned spruces still dispersed viable seeds 8 years after fire. Wein (1975) reported that the presence of a few viable seeds 20 years after fire in a forest–tundra stand near Inuvik, Northwest Territories. If one considers that black spruce seeds are dispersed within a short distance, i.e., about two to three times the height of dominant trees (Heinselman 1957), and that buried viable seeds are absent at such sites (Johnson 1975; Payette *et al.* 1982), it is likely that the major seed source for postfire regeneration comes from burned trees. Therefore, any factor modifying the seed source will affect the pattern of

postfire establishment, as shown at site 3 (Fig. 6) where the high density of postfire conifers was caused by living trees shedding numerous seeds over a very small burned area (5000 m²). The presence at this site of several tamaracks supports this conclusion. Tamarack is ill-adapted to fire, and postfire regeneration proceeds generally from nearby living trees.

Data on black spruce seed dispersal after fire indicate that both number and viability of seeds decrease with time (Wilton 1963; Zasada *et al.* 1979). However, our age structure data (Fig. 6) show a regeneration peak occurring, in general, between 5 and 14 years after fire, a delay also observed by Hatcher (1963) and Sirois (1988). The low success in spruce establishment immediately after fire may be closely associated with adverse soil conditions. Water stress may be detrimental to germinating seeds and seedlings (Black and Bliss 1980), as well as extreme temperature conditions prevailing in recently burned areas (Rouse and Kershaw 1971; Rouse 1976; Kershaw 1977; Viereck and Foote 1979). It seems likely that the regeneration peak occurring 5–14 years after fire could be related to the phasing out of such lethal conditions, whereas the end of the active regeneration period would coincide to depletion of the seed source in burned trees and development of a lichen cover of sufficient density to prevent seedling establishment from mature postfire spruces. The youngest spruces able to produce cones were about 25 years old. At site 5 (Fig. 6), where a second period of seedling establishment occurred 15–20 years after the first regeneration peak, the presence of tamarack suggested the incidence of an external seed source. In old-growth woodlands, the thick, continuous lichen cover constituted an important physical barrier to seed germination, as it restricts seed penetration into the soil surface. This explains why most seedlings were confined to microhabitats such as lichen polygons and exposed mineral substrata associated with wind-thrown trees. Therefore, we can conclude that spruce populations must establish during a short period after fire (20–30 years) for reconstruction of lichen woodlands.

During a long fire-free period, layering became the dominant regenerative process. Several layers have reached tree size (7–8 m high) similar to postfire dominants, but their biomass was usually lower than that of the mother plant. Increase in stem density associated with layering was accompanied, however, by death of dominant trees; this explains why the overall spruce cover remained more or less constant through time (Fig. 3o). The similar age structure of layers in 129- and 250-year-old forests (Fig. 6, sites 7 and 8), including many young layers but few old ones, suggests high mortality of young layers.

Soil conditions

According to uniform thickness of the FH horizon whatever the site, only light fires occurred in the study area. Viereck and Schandelmeyer (1980) suggested that nutrient availability increases after fire. In northern Saskatchewan, where severe fires are frequent, Scotter (1964) reported higher pH and Ca and P contents in soils of recently burned sites, whereas total N and exchangeable Na, K, and Mg remained rather constant. At Schefferville, Moore (1980) showed an increase in pH and Ca availability after fire in soils of lichen–spruce woodlands. However, this amelioration in soil conditions was of short duration (<5 years) and of low amplitude compared with that of other ecosystems.

The analysis of all chemical factors considered in this study (Fig. 8) suggested that the intersite variability could be attributed more to spatial variability than to changes during succession. Soil sampling here was not designed to demonstrate any fire influence on edaphic conditions. On a regional basis, however, it seemed that spatial variability of soil conditions was greater than temporal variability. This indicated that fire had little influence on the soil conditions reported here. The incidence of light fires and regional dominance of open spruce forests with low biomass and low nutrient content of the lichen thalli suggest that fire had no significant effect on the nutrient status of lichen-woodland soils. In this respect, changes in microclimatic conditions associated with fire (Rouse and Kershaw 1971; Rouse 1976; Kershaw 1977) would be more influential.

Long-term development of the lichen-spruce woodland

In North America, lichen-spruce woodlands are the dominant forest type in dry and mesic sites of boreal and subarctic regions. These open forests are generally associated with recurrent fires (Auclair 1983; Viereck 1983), leading some authors to hypothesize that their long-term maintenance requires periodic fire disturbances (Rowe *et al.* 1975 cited in Auclair 1983).

The fire regime in the Grande rivière de la Baleine area, characterized by a 100-year rotation period (Payette *et al.* 1989), limits the development of old-growth forests. However, no moss forest was found on well-drained sites, and the oldest stands (about 250 years old) were forming typical lichen-spruce woodlands. As a dominant moss species in closed forests, *Ple. schreberi* is restricted in old woodlands to shaded areas under spruce clones. Because the development of a moss cover depends on closure of the tree cover, the absence of *Pleurozium*-spruce forests could be attributed to rarity of old fire-free sites. However, our data do not support this hypothesis because the 129- and 250-year-old woodlands showed the same structural and vegetational characteristics. At the time the *Cladina stellaris* cover was well developed, no major change was recorded in floristic composition, vegetation structure, diversity, and, above all, spruce cover. Thus, the presence of a continuous lichen carpet appeared to be the main factor for persistence of the lichen-spruce facies in absence of fire. Suitable microsites for seed germination are too spatially restricted for progressive closure of the tree canopy. Also, the typically slow clonal development occurring in lichen-spruce woodlands cannot compensate for stem mortality. That is why both regenerative processes in old-growth woodlands, i.e., sporadic sexual regeneration and layering, can only maintain a low spruce cover.

The long-term persistence of lichen-spruce woodlands has been demonstrated at the forest limit in northern Quebec, where 800- to 900-year-old postfire woodlands were recently analyzed (Payette *et al.* 1985). Although these woodlands developed under harsher climatic conditions, a similar developmental structure, with sporadic seed regeneration and extensive layering, occurred during the past centuries. In southeastern Labrador, succession from lichen-spruce woodland to *Pleurozium*-spruce forest was most likely associated with a more humid and milder climate (Foster 1985). The Labrador climate induced a different fire regime with a longer fire rotation period (Foster 1983), which probably facilitated moss growth and tree canopy expansion. According to these

data, it is highly improbable that lichen-spruce woodlands growing on well-drained soils, under the conditions of continental climate as reported in the study area, could be replaced by closed-crown forests during succession as suggested by Kershaw (1977, 1978).

Conclusion

The postfire recovery sequence leading to the development of *Cladina stellaris* - *Picea mariana* woodlands in the Grande rivière de la Baleine area is broadly similar to other chronosequence reported elsewhere in Canada, i.e., (i) fast reestablishment and absence of vascular species replacement during succession and (ii) dominance of lichens. Most changes in floristic composition and vegetation development occurred during the first 100 years of succession. In vascular plants, the changes were associated with growth and development of shrub species and black spruce. Species interactions during succession appeared more important in lichens and mosses, as the most important changes referred to species abundance most likely controlled by differential growth rates and competition. The postfire chronosequence was also characterized by important changes in species diversity during the first 65 years of postfire succession.

Black spruce regeneration from seeds occurred almost exclusively during the first years of vegetation recovery, whereas layering became the dominant regenerative process between major fire-disturbance periods. Most fires recorded in the area were of low severity and had no significant influence on soil conditions. The lichen-spruce woodlands can persist during a long fire-free period, at least 250 years, although such conditions were unfrequent in the study area. The present status of black spruce regeneration in old-growth woodlands suggests that maintenance of an open tree canopy as time passed is the rule, and lichen woodlands can perpetuate in absence of fire.

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